

Chapter 15

Connecting LANs, Backbone Networks, and Virtual LANs

15-1 CONNECTING DEVICES

In this section, we divide connecting devices into five different categories based on the layer in which they operate in a network.

Topics discussed in this section:

Passive Hubs

Active Hubs

Bridges

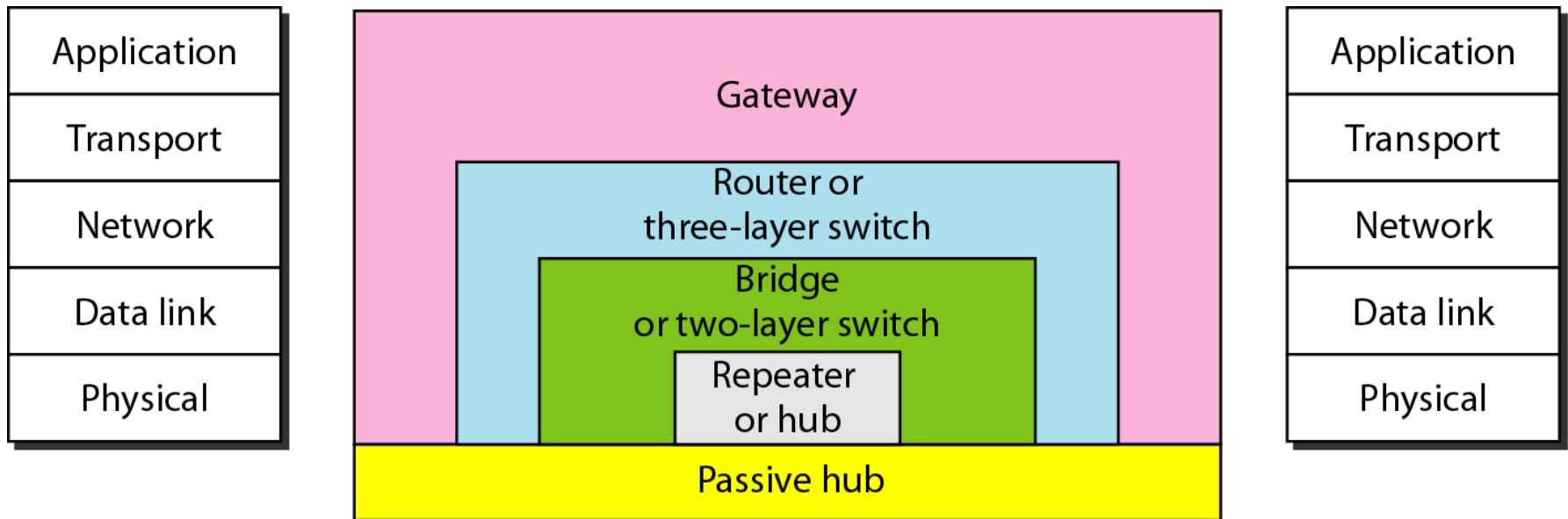
Two-Layer Switches

Routers

Three-Layer Switches

Gateways

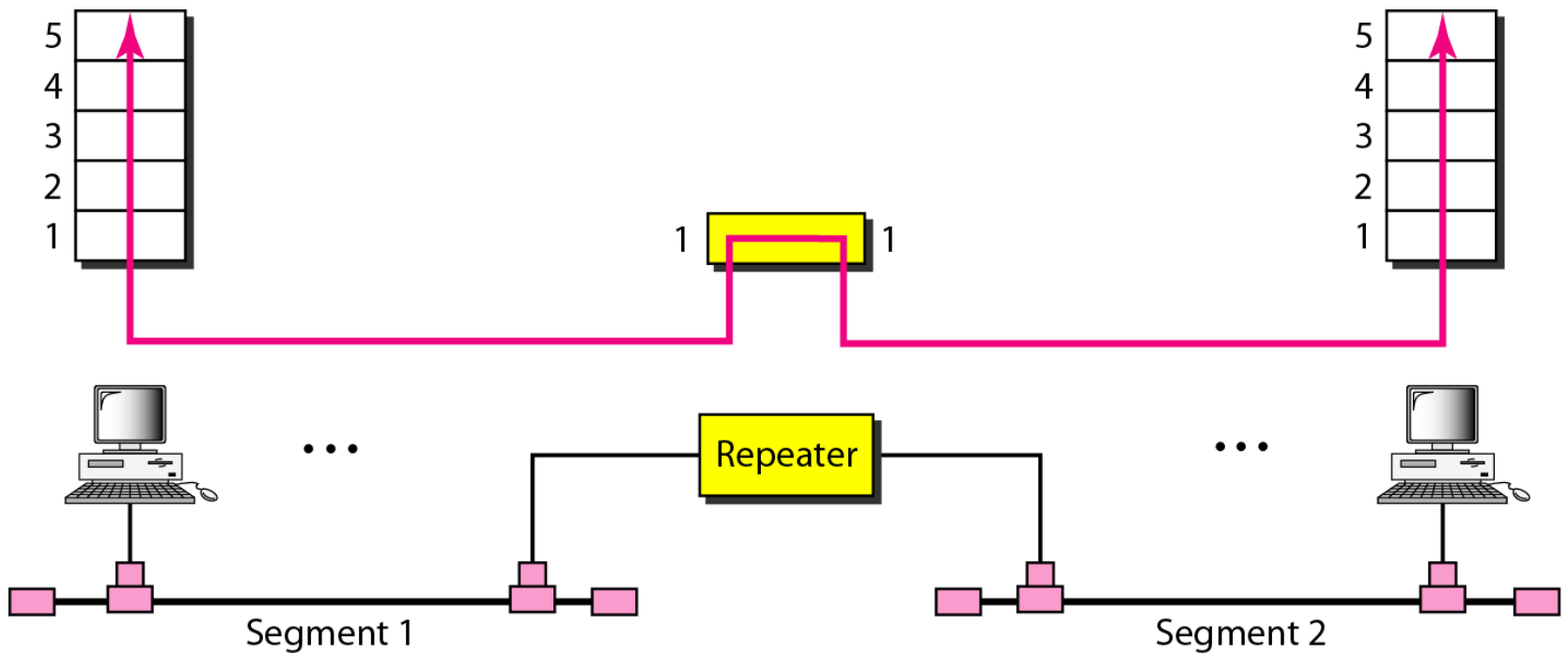
Figure 15.1 *Five categories of connecting devices*



Difference between Passive Hub and Hub

Feature	Passive Hub	Repeater
Power Requirement	No	Yes
Functionality	Simple connector	Central repeater
Distance Support	Short	Longer
OSI Layer	Physical Layer (L1)	Physical Layer (L1)
Use Case	Small networks	Older Ethernet LANs

Figure 15.2 *A repeater connecting two segments of a LAN*





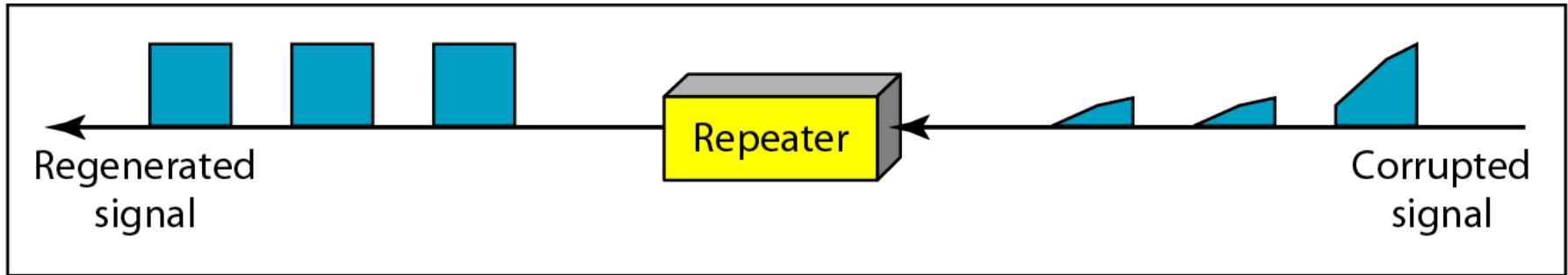
Note

A repeater connects segments of a LAN.

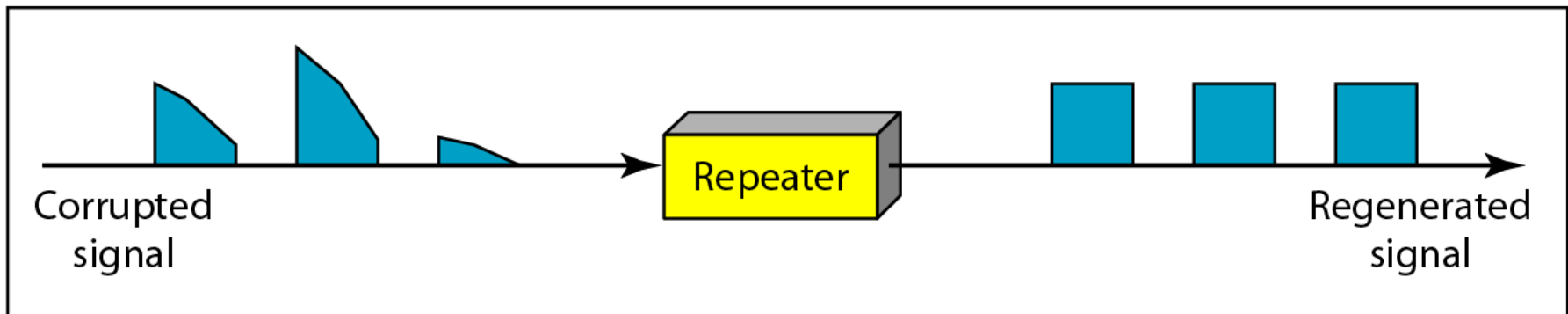
**A repeater is a regenerator,
not an amplifier.**

**A repeater is a regenerator,
not an amplifier.**

Figure 15.3 *Function of a repeater*

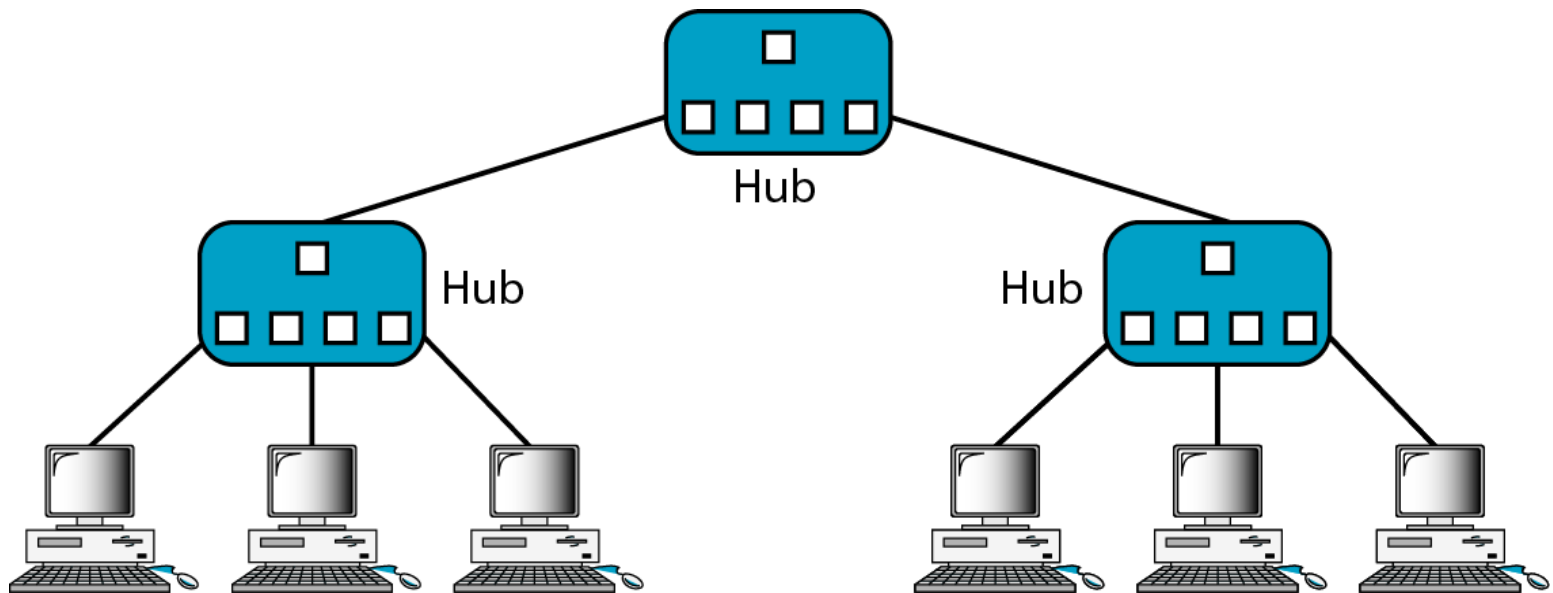


a. Right-to-left transmission.



b. Left-to-right transmission.

Figure 15.4 *A hierarchy of hubs*

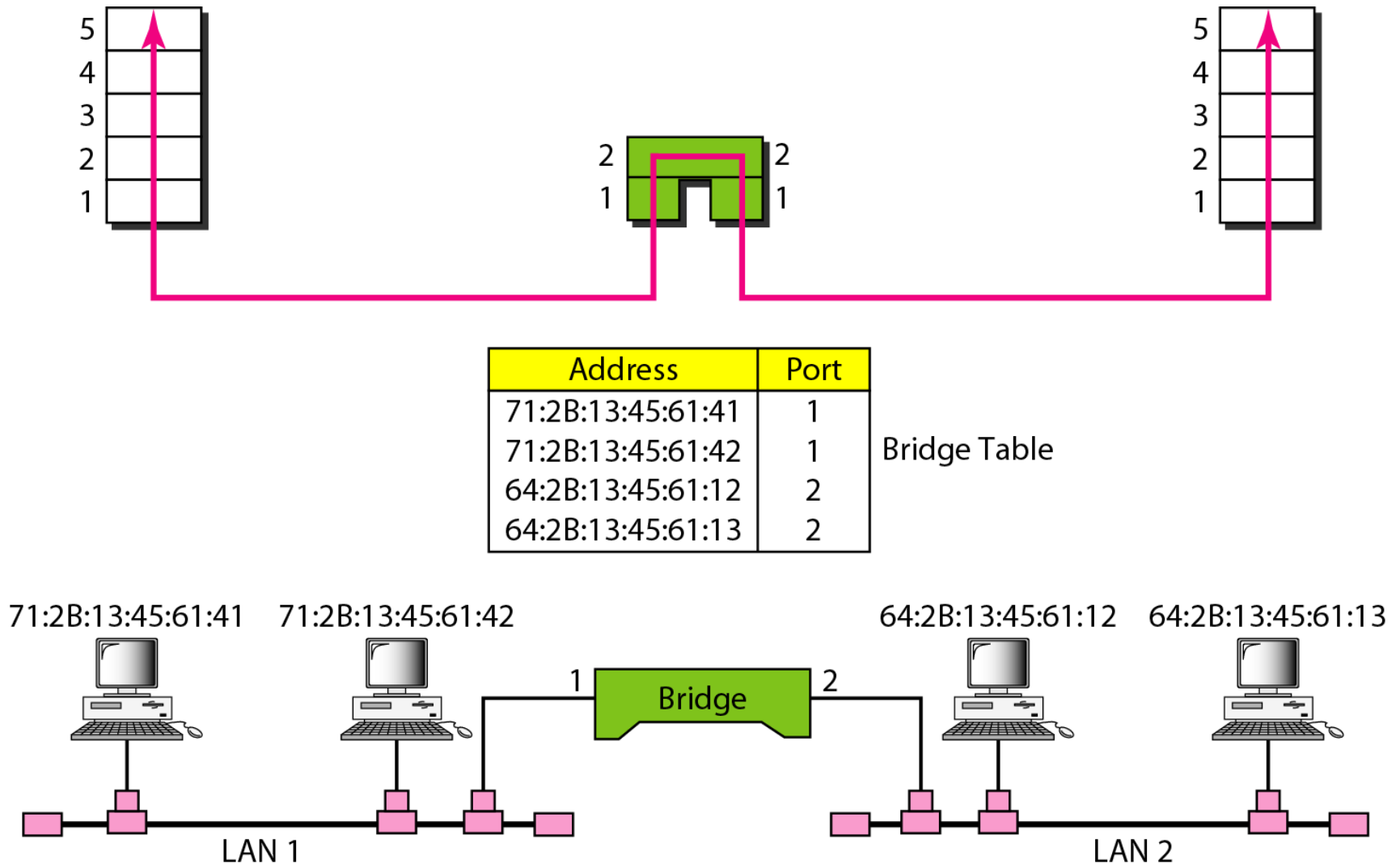




Note

**A bridge has a table used in
filtering decisions.**

Figure 15.5 *A bridge connecting two LANs*





Note

A bridge does not change the physical (MAC) addresses in a frame.

Figure 15.6 *A learning bridge and the process of learning*

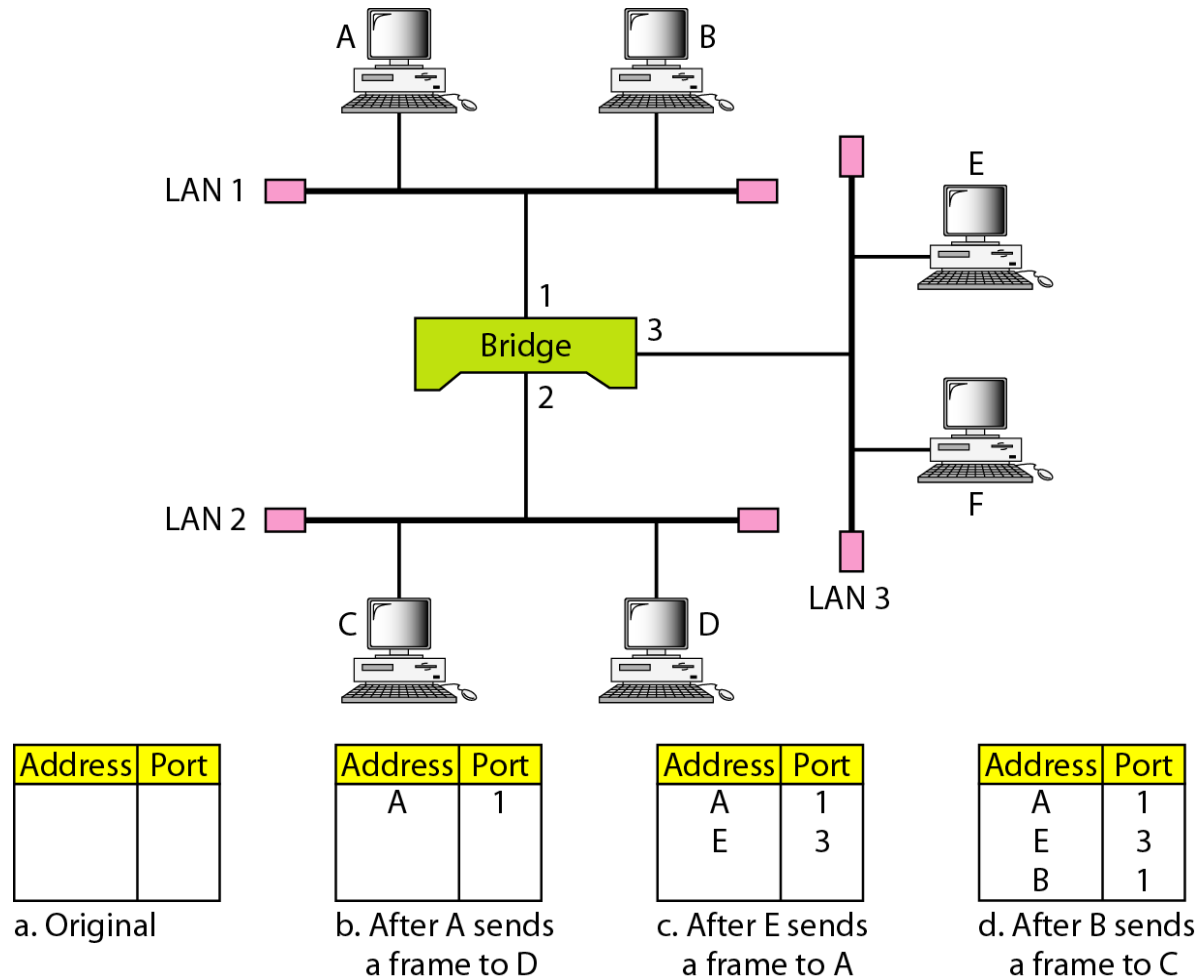
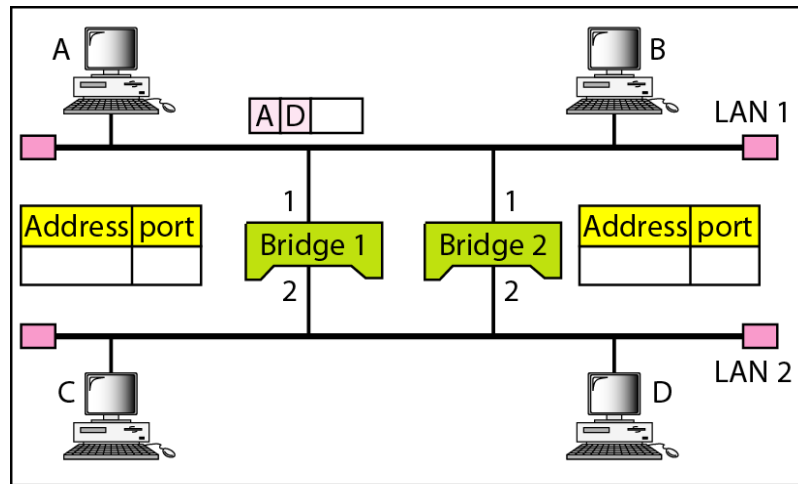
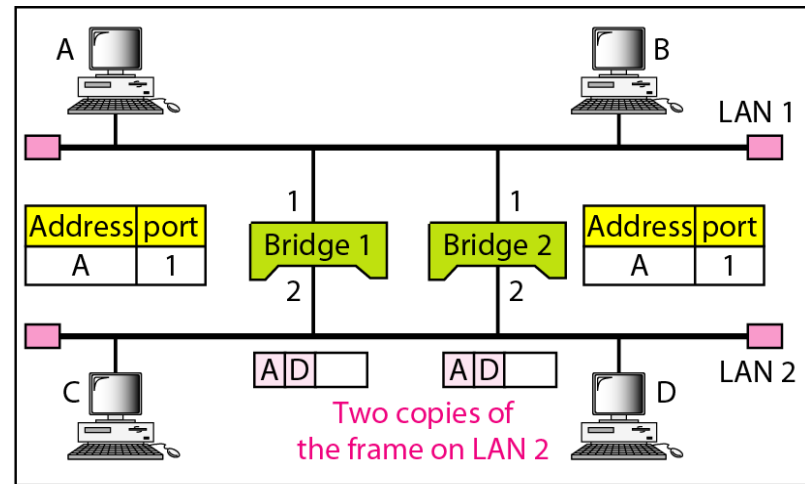


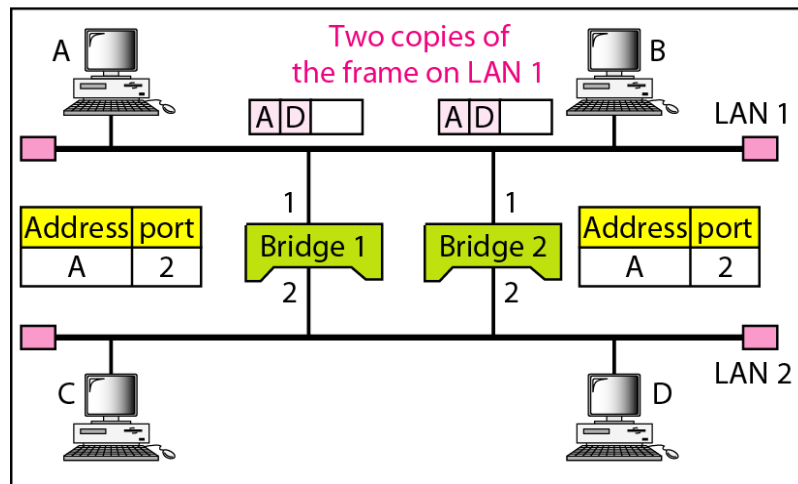
Figure 15.7 *Loop problem in a learning bridge*



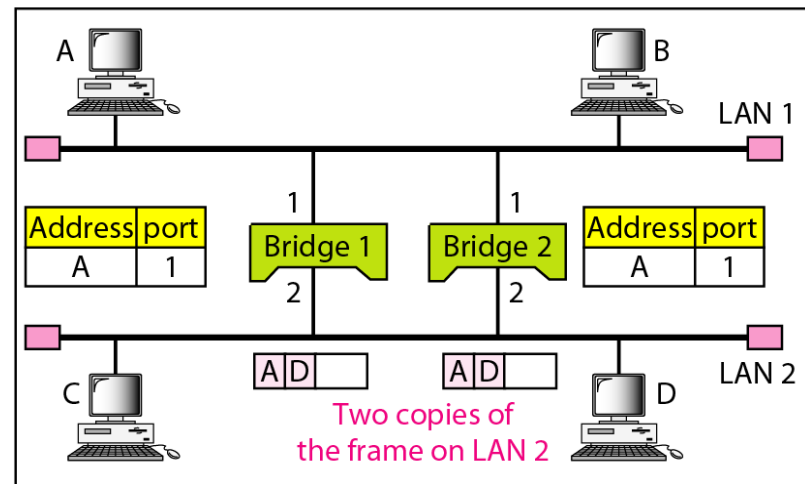
a. Station A sends a frame to station D



b. Both bridges forward the frame

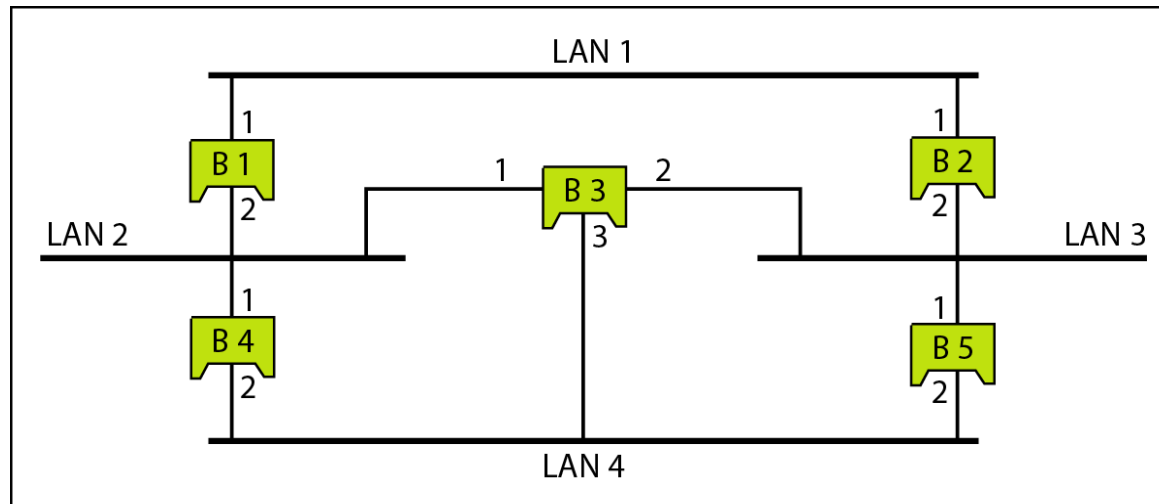


c. Both bridges forward the frame

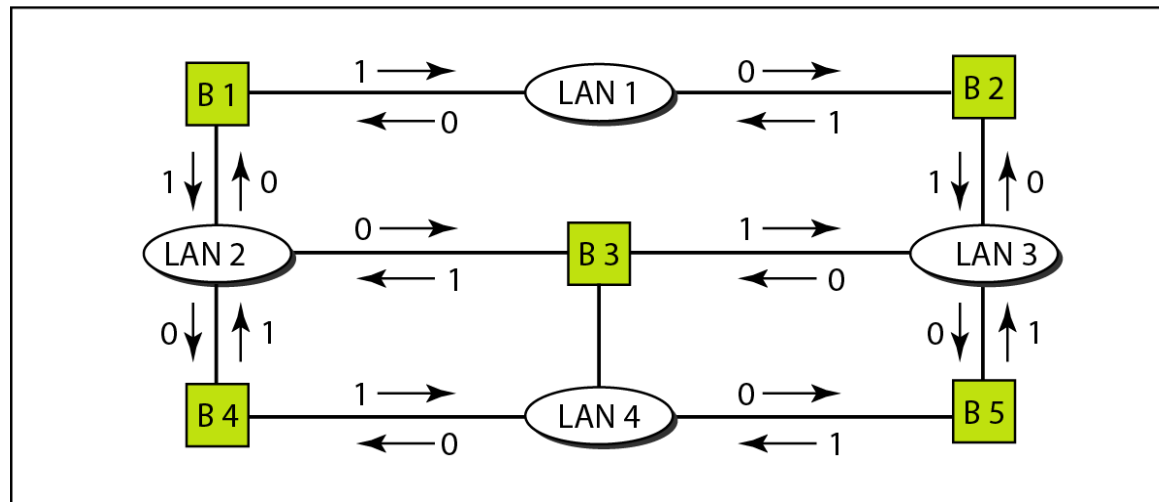


d. Both bridges forward the frame

Figure 15.8 *A system of connected LANs and its graph representation*

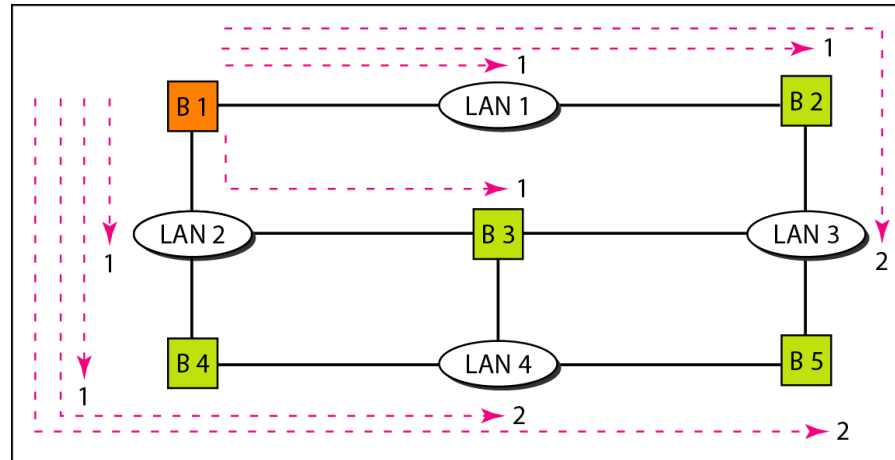


a. Actual system

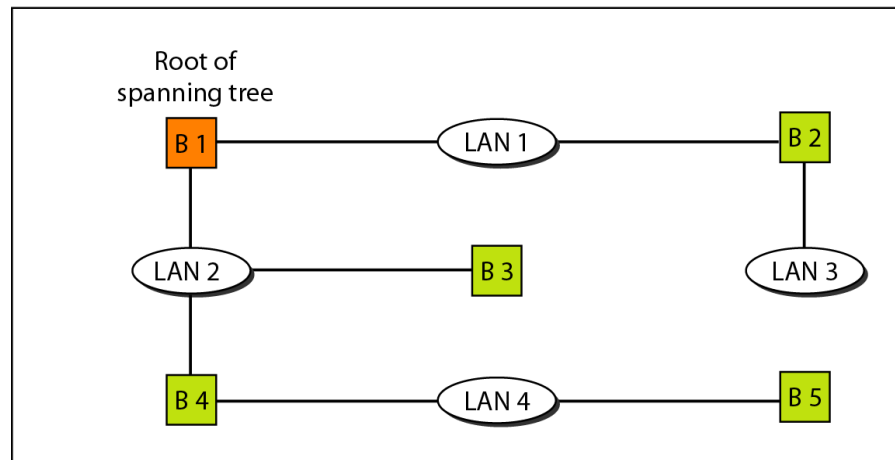


b. Graph representation with cost assigned to each arc

Figure 15.9 *Finding the shortest paths and the spanning tree in a system of bridges*

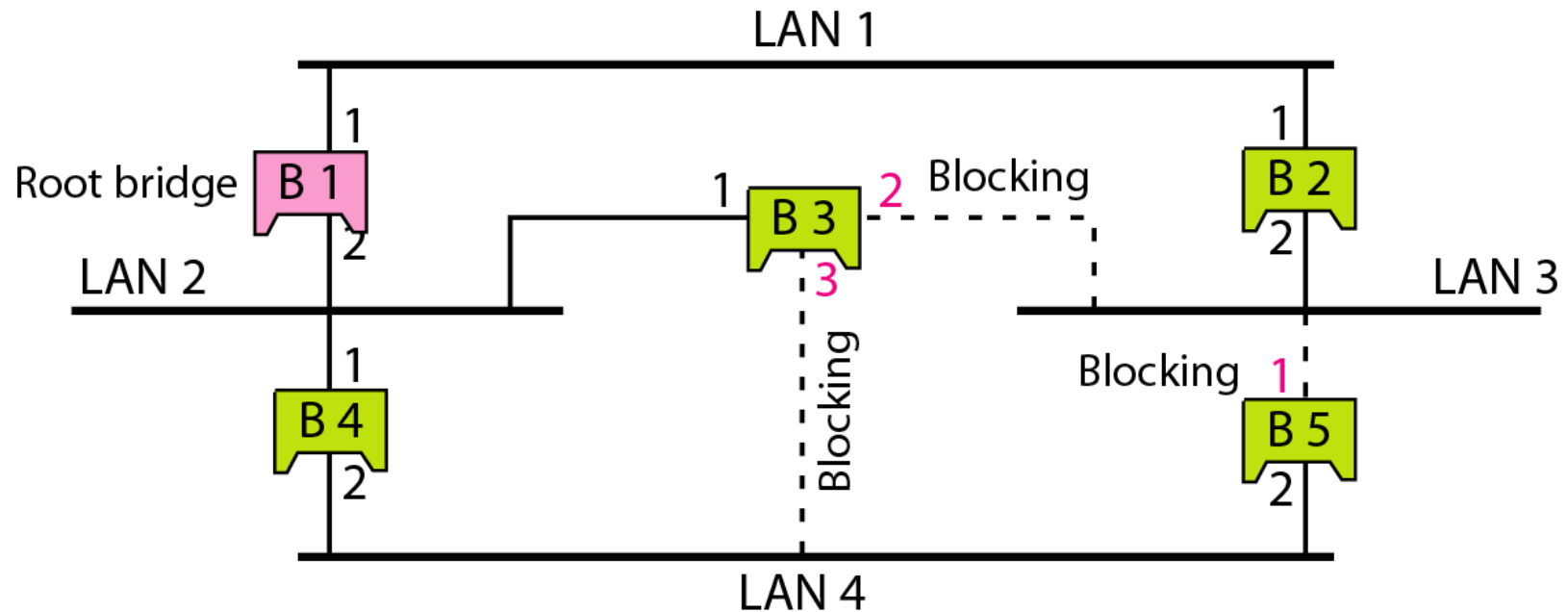


a. Shortest paths



b. Spanning tree

Figure 15.10 *Forwarding and blocking ports after using spanning tree algorithm*

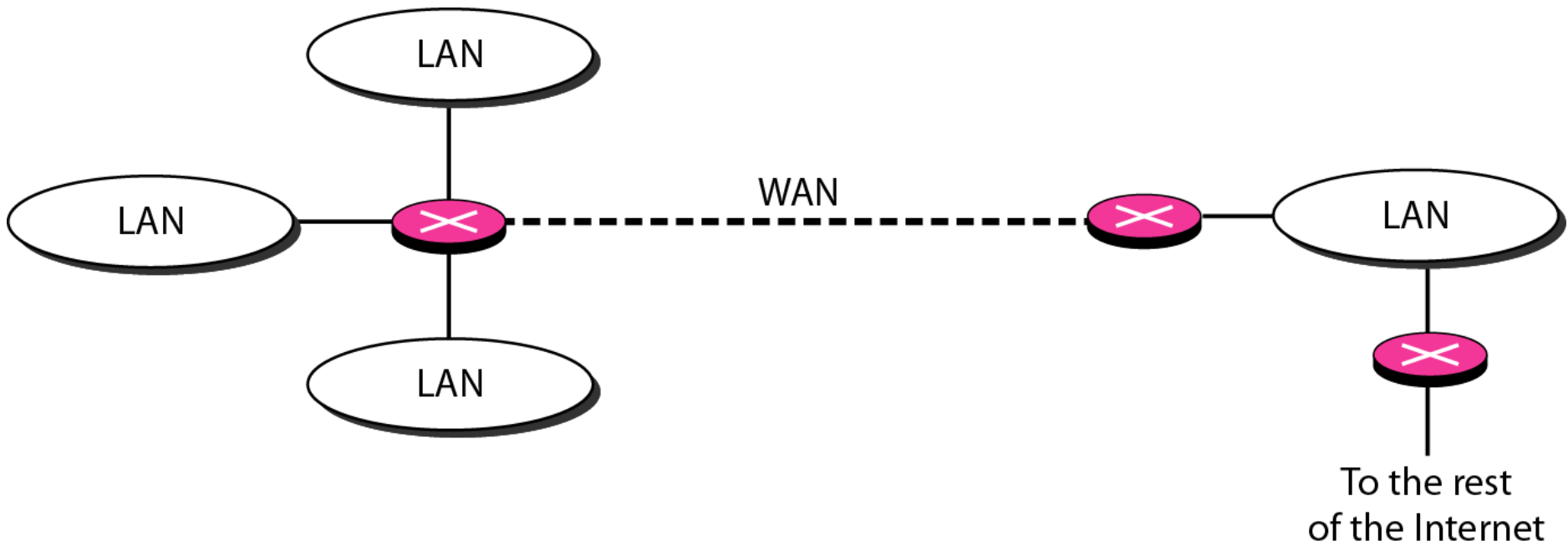


Ports 2 and 3 of bridge B3 are blocking ports (no frame is sent out of these ports). Port 1 of bridge B5 is also a blocking port (no frame is sent out of this port).

Bridges

- A bridge connects two or more LAN segments.
 - Operates at Data Link Layer (Layer 2).
 - Forwards frames based on MAC addresses.
 - Filters traffic to reduce unnecessary broadcasts.
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- Filtering: Discards frames if destination is on the same segment.
 - Forwarding: Sends frame to another segment if needed.
 - Learning: Builds a forwarding table dynamically using source MAC addresses.

Figure 15.11 *Routers connecting independent LANs and WANs*



15-2 BACKBONE NETWORKS

A backbone network allows several LANs to be connected. In a backbone network, no station is directly connected to the backbone; the stations are part of a LAN, and the backbone connects the LANs.

Topics discussed in this section:

Bus Backbone

Star Backbone

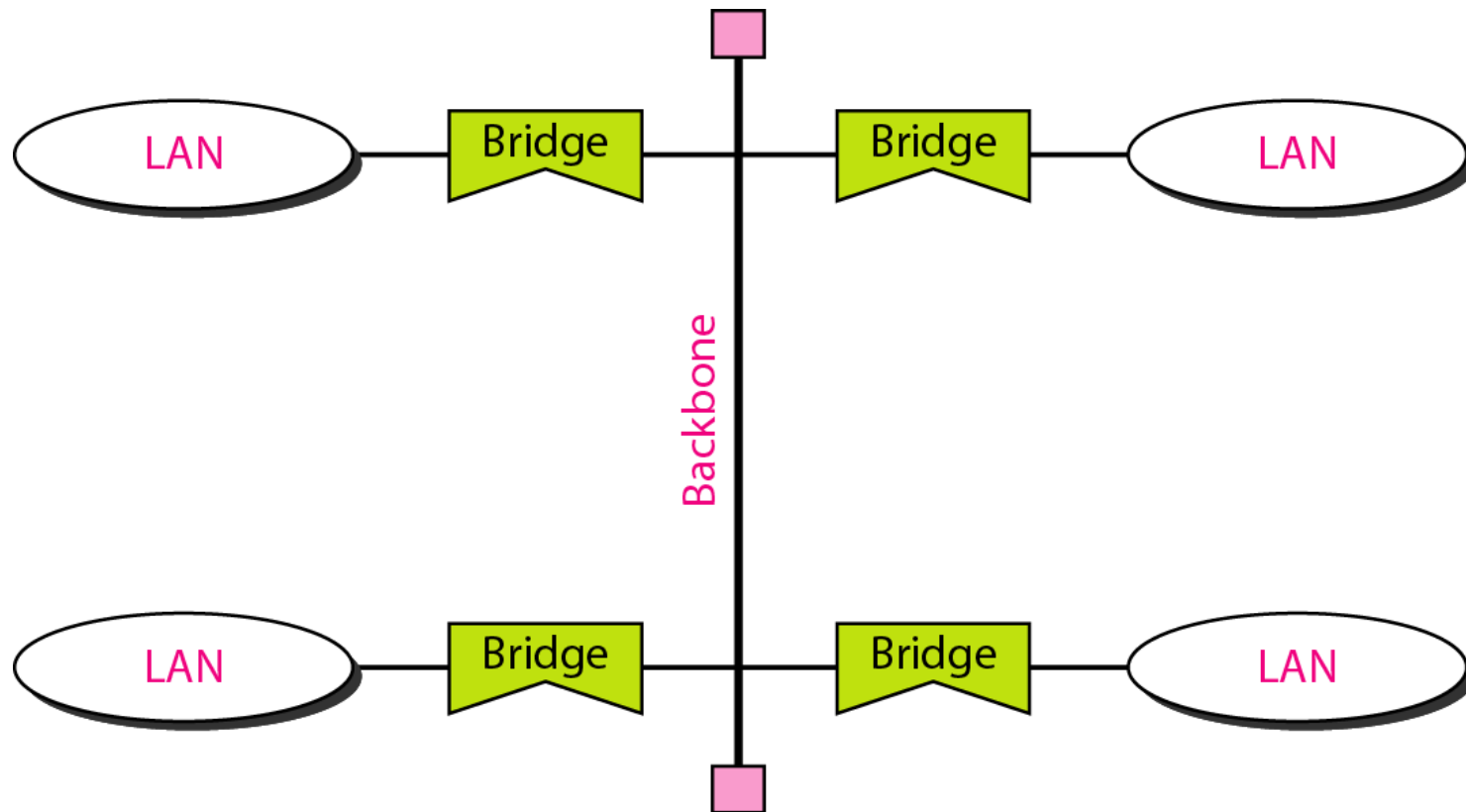
Connecting Remote LANs



Note

**In a bus backbone, the topology
of the backbone is a bus.**

Figure 15.12 *Bus backbone*



Note

**In a star backbone, the topology of the backbone is a star;
the backbone is just one switch.**

Figure 15.13 *Star backbone*

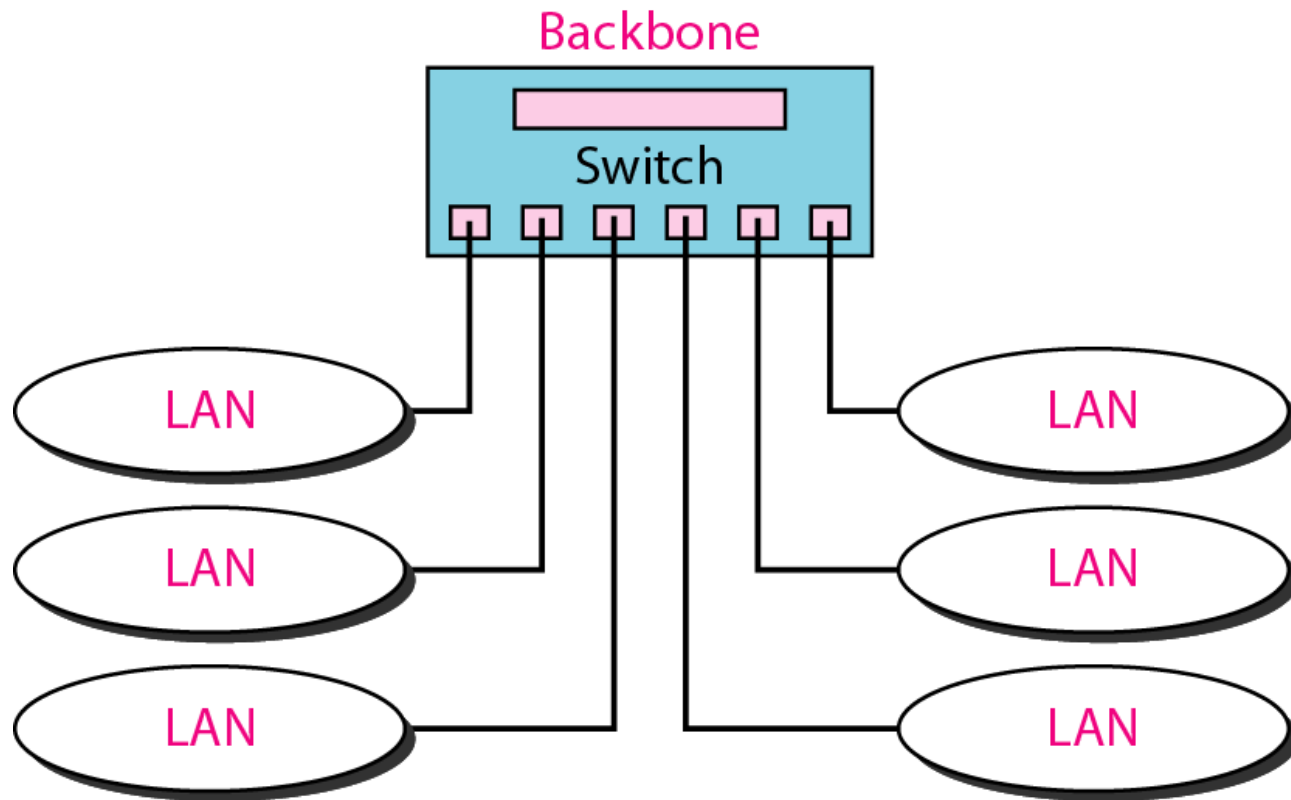
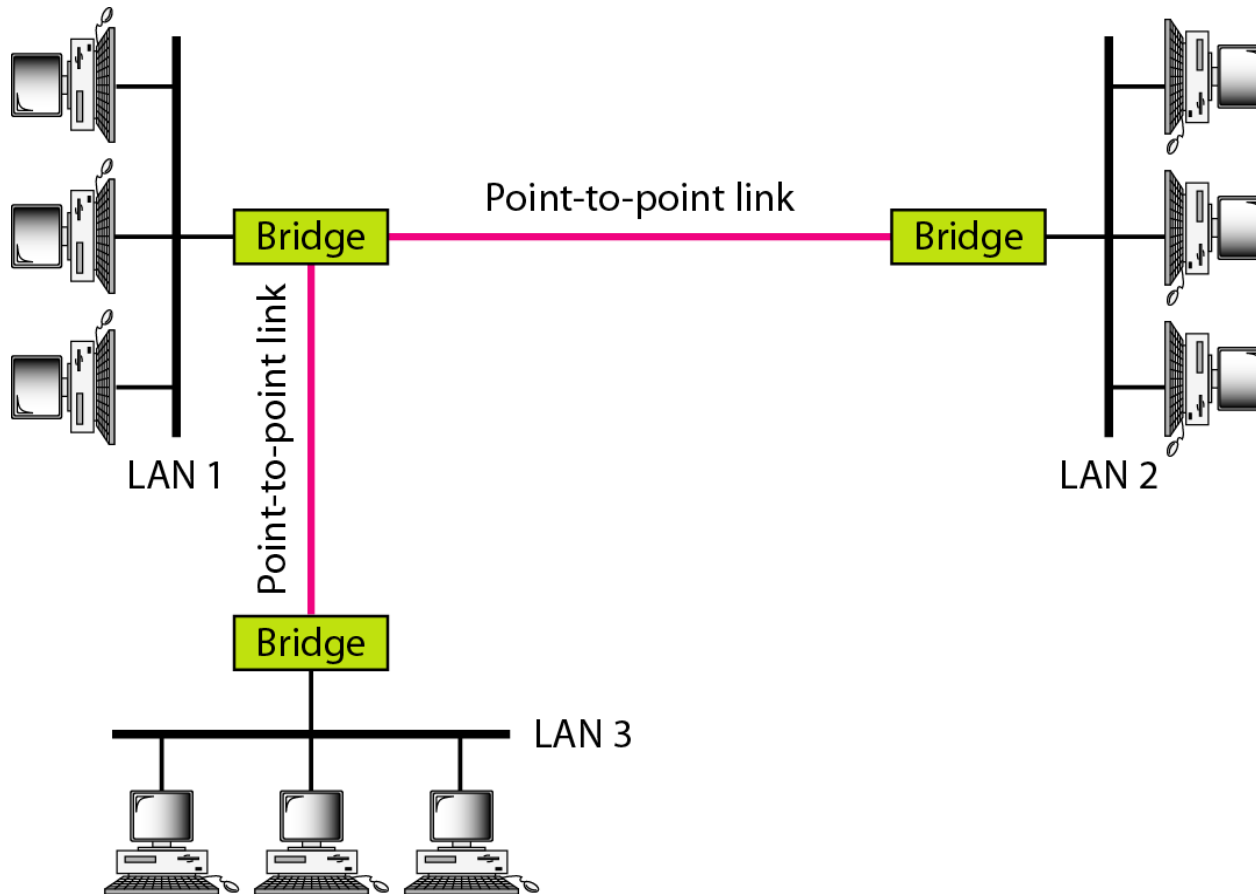


Figure 15.14 *Connecting remote LANs with bridges*





Note

A point-to-point link acts as a LAN in a remote backbone connected by remote bridges.

15-3 VIRTUAL LANs

*We can roughly define a **virtual local area network (VLAN)** as a local area network configured by software, not by physical wiring.*

Topics discussed in this section:

Membership

Configuration

Communication between Switches

IEEE Standard

Advantages

Figure 15.15 *A switch connecting three LANs*

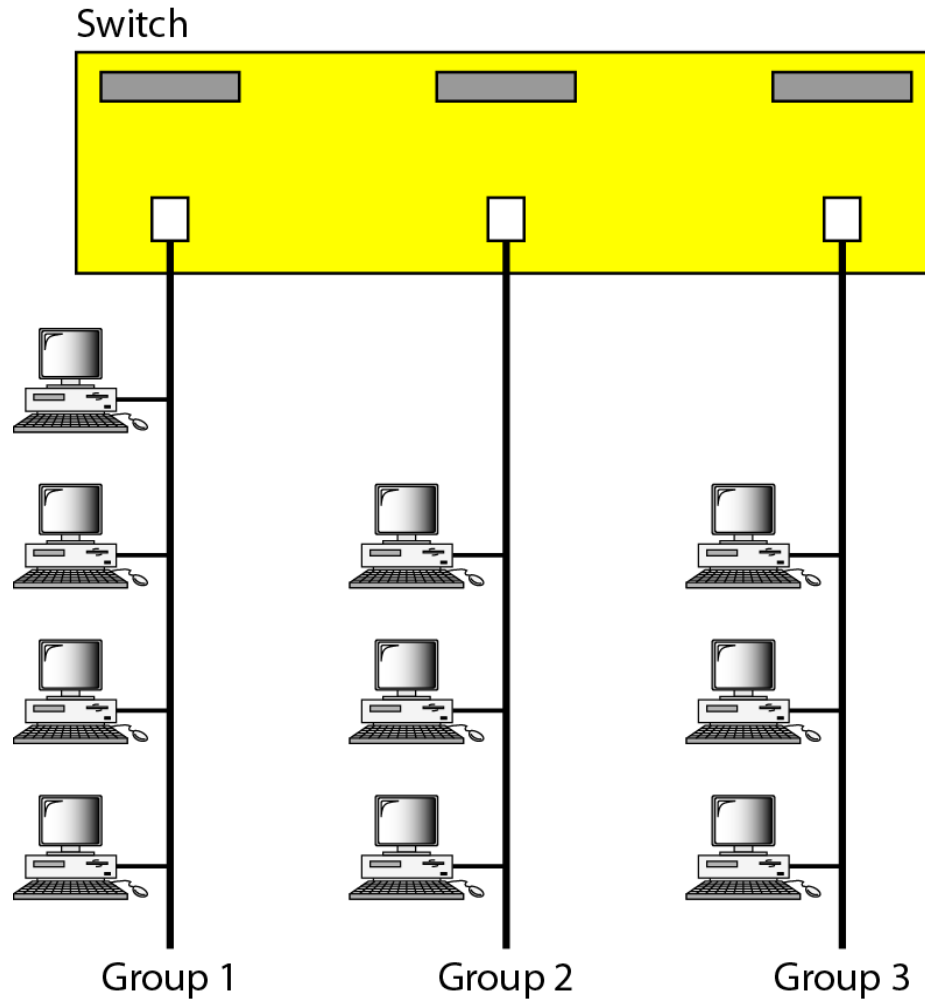


Figure 15.16 *A switch using VLAN software*

Switch with VLAN software

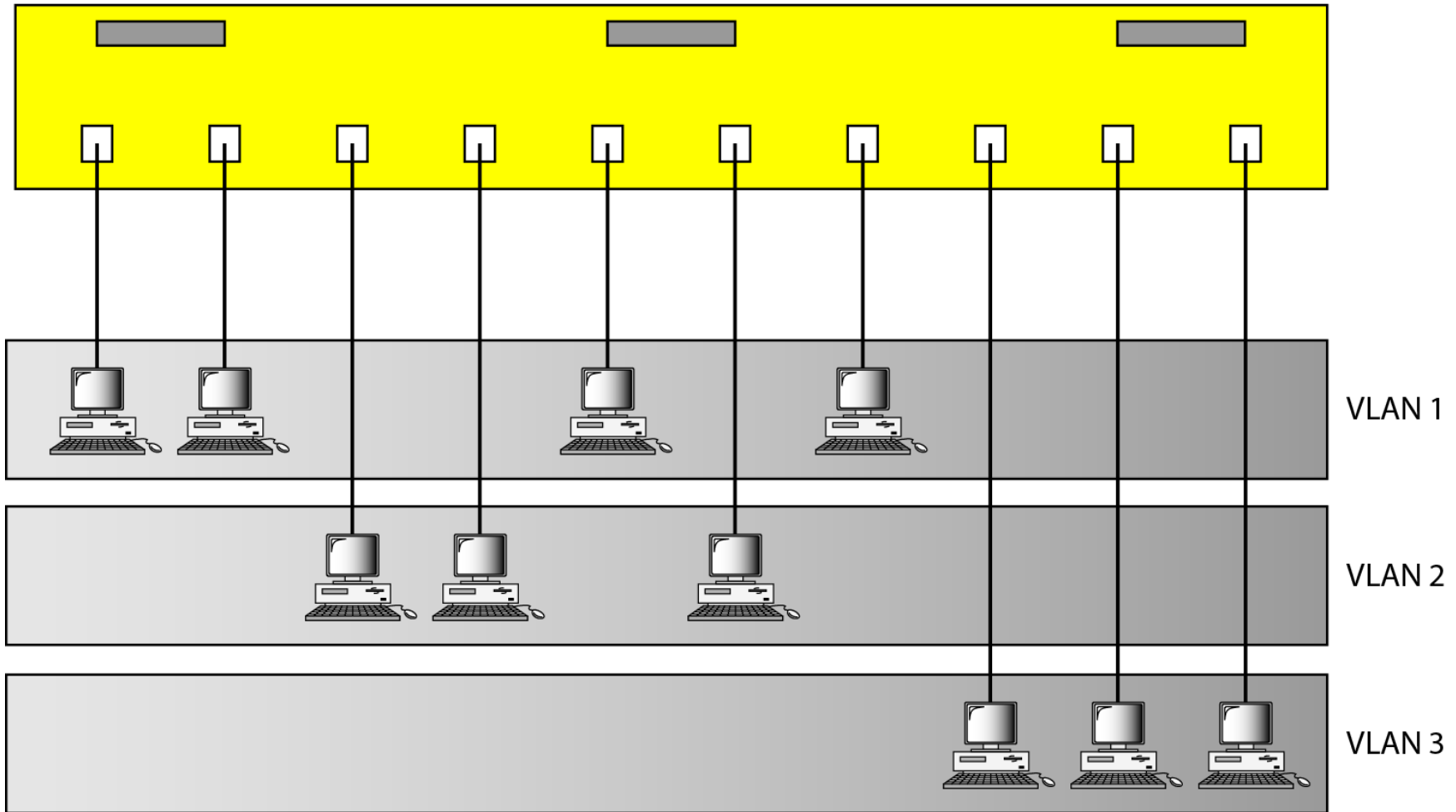
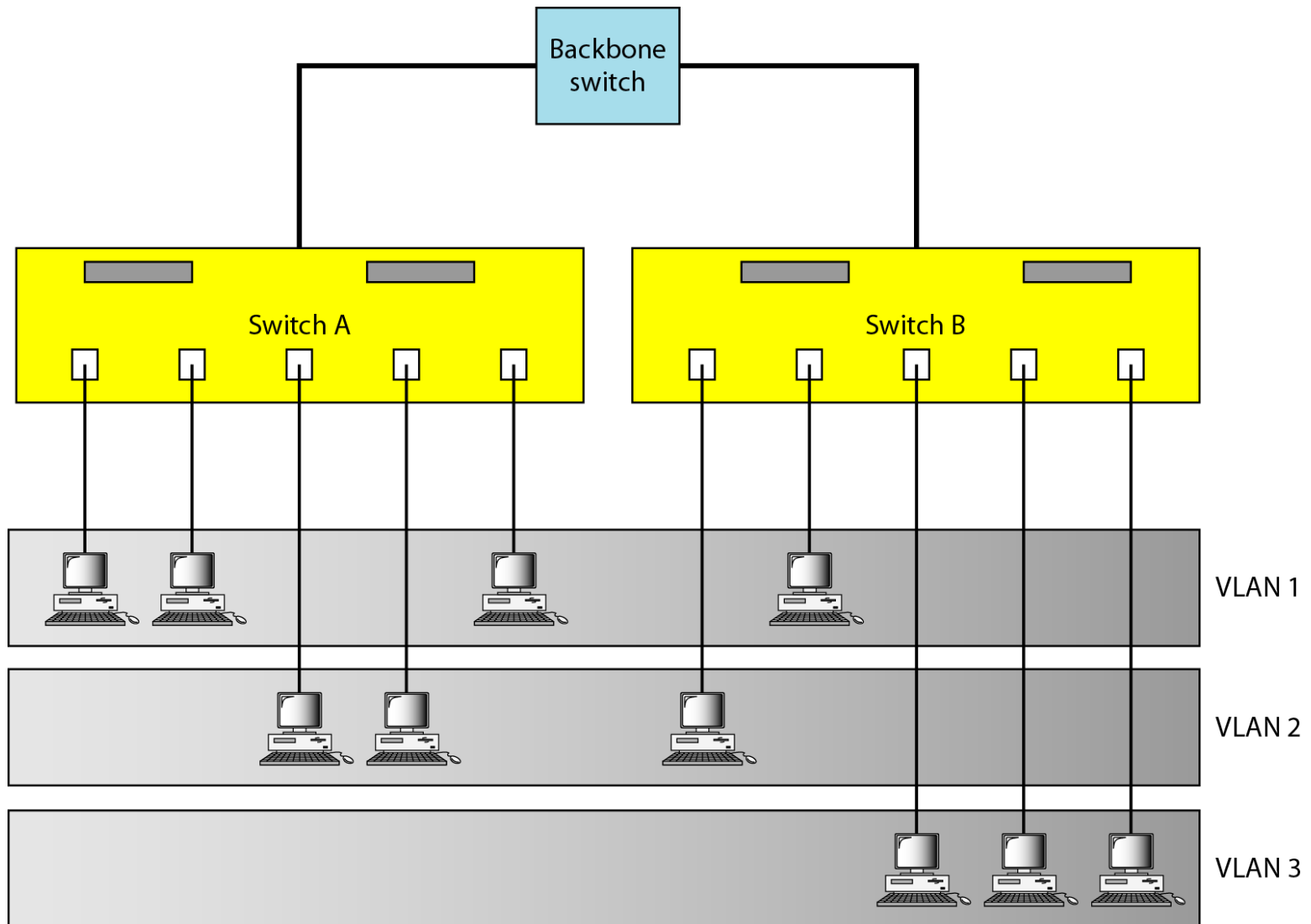


Figure 15.17 *Two switches in a backbone using VLAN software*





Note

VLANs create broadcast domains.